

Year 2 Assessed Problems

Semester 2

Assessed Problems

SOLUTIONS TO BE SUBMITTED

ON CANVAS BY

Wednesday 25 February

at 17:00

Lagrangian & Hamiltonian Mechanics

1. A Lagrangian for a spinning top can be written as

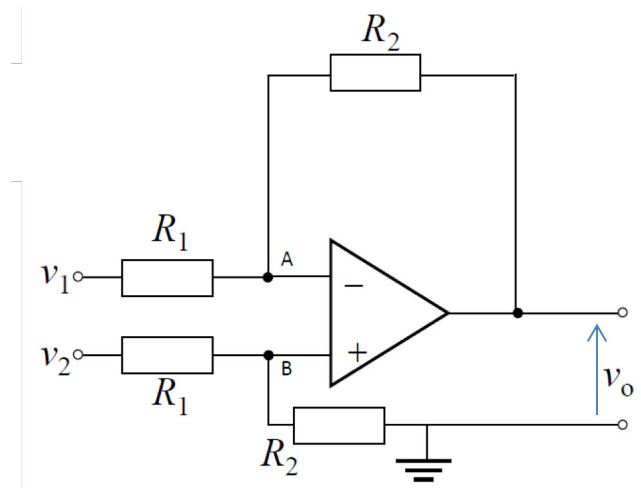
$$L = \frac{I_1}{2} (\dot{\phi}^2 \sin^2 \theta + \dot{\theta}^2) + \frac{I_3}{2} (\dot{\phi} \cos \theta + \dot{\psi})^2 - mgh \cos \theta,$$

where θ , ϕ and ψ are the so called Euler angles, I_3 is the moment of inertia about the axis of symmetry, I_1 the moment of inertia about a perpendicular axis, and h the position of the centre of mass along the axis of symmetry.

- (a) Identify the cyclic variables for the system. Derive expressions for the conjugate momenta p_θ , p_ϕ and p_ψ . Which of these momenta are conserved? [4]
- (b) Derive an expression for the Hamiltonian of the system. Explain why it is conserved. [3]
- (c) Use all the conserved quantities to reduce the problem to that with one degree of freedom, θ , i.e. derive the equation of motion in terms of θ only. [3]

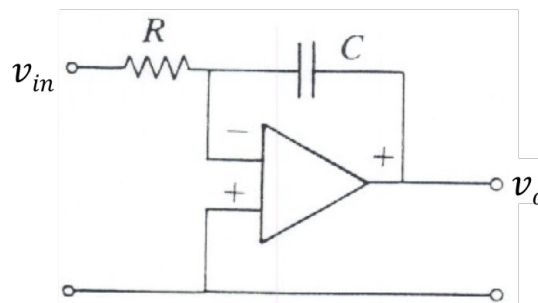
Operational amplifiers

1. The circuit below is a difference amplifier. Assuming an ideal op-amp, find the voltage v_o in terms of $v_2 - v_1$.



[3]

2. Consider the integrator shown below. Assume an ideal op-amp. Let $R = 1 \text{ M}\Omega$ and $C = 10 \text{ nF}$.



- (a) Show that in the time domain v_o is proportional to $\int v_{in}$, and find the constant of proportionality.

[3]

- (b) Consider a pulse waveform at the input with amplitude $A = 1 \text{ V}$ and width $T = 0.1 \text{ ms}$. Draw the input and output voltages against time, labelling the values on the axes.

[1]

- (c) Calculate the closed loop gain of the circuit and sketch the corresponding Bode plots for magnitude and phase.

[3]

Year 2: Observational Astronomy

Assessed Problem Sheet #2

1. Starting from the following definition of the AB magnitude system

$$m = -2.5 \log S_\nu - 56.1,$$

derive an expression for the arrival rate of photons per square metre per second in terms of m , the effective spectral resolution R , and the Planck constant h . State any assumptions that you make.

[3 marks]

2. Estimate the amount of observing time required to detect a source with apparent AB magnitude $m = 30$ at a signal-to-noise ratio of $SNR = 10$ with a telescope of diameter $D = 10\text{m}$. You may assume that the only source of noise is the Poisson uncertainty on the arrival rate of photons, and that all of the photons that arrive are detected. **[3 marks]**
3. The astronomer measures the spectral flux density of a star at a wavelength of $\lambda_{\text{observed}} = 550\text{nm}$ to be $S_\lambda = 5 \times 10^{-18} \text{ W m}^{-2} \text{ nm}^{-1}$. What is the spectral flux density of this star in units of $\text{erg s}^{-1} \text{ cm}^{-2} \text{ \AA}^{-1}$? **[2 mark]**
4. The spectral flux density of Vega at 550nm is $S_\lambda = 3.6 \times 10^{-9} \text{ erg s}^{-1} \text{ cm}^{-2} \text{ \AA}^{-1}$. What is the apparent magnitude of the star in part (3) at 550nm ? **[2 marks]**

Optics Assessed Problem sheet 2

- 1 Show that the paraxial ray transformation matrix for a curved dielectric interface with radius of curvature R is given by (with clear labelling and sketch) [10]

$$\begin{pmatrix} 1 & 0 \\ \frac{1}{R} \left(\frac{n_1}{n_2} - 1 \right) & \frac{n_1}{n_2} \end{pmatrix}$$

